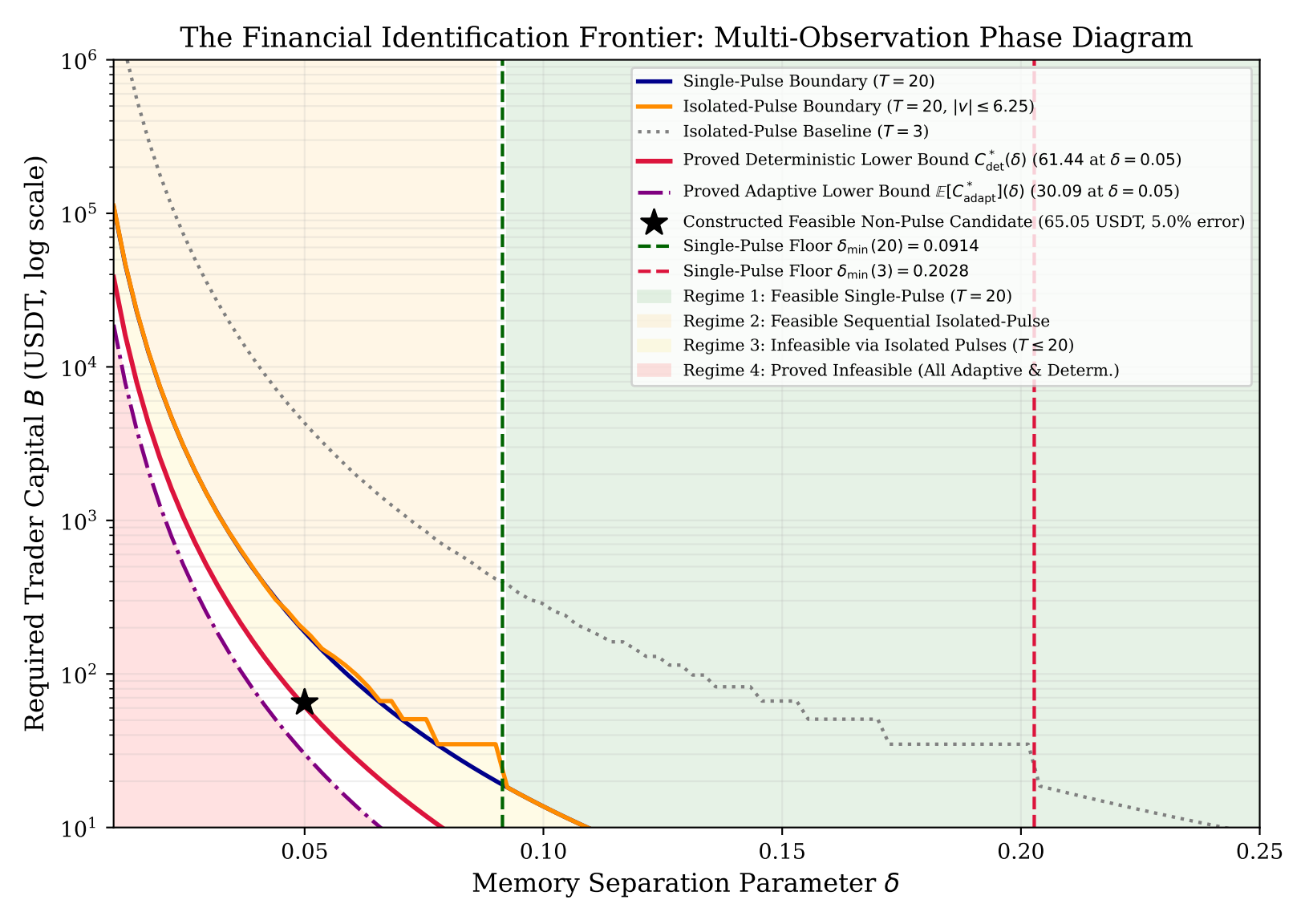
The Financial Identification Frontier: Multi-Observation Phase Diagram
Required Trader Capital B (USDT, log scale)
Memory Separation Parameter δ
10^6
10^5
10^4
10^3
10^2
10^1
0.05
0.10
0.15
0.20
0.25
Single-Pulse Boundary ($T = 20$)
Isolated-Pulse Boundary ($T = 20$, $|\nu| \leq 6.25$)
Isolated-Pulse Baseline ($T = 3$)
Proved Deterministic Lower Bound $C^*_{\text{det}}(\delta)$ (61.44 at $\delta = 0.05$)
Proved Adaptive Lower Bound $\mathbb{E}[C^*_{\text{adapt}}](\delta)$ (30.09 at $\delta = 0.05$)
Constructed Feasible Non-Pulse Candidate (65.05 USDT, 5.0% error)
Single-Pulse Floor $\delta_{\min}(20) = 0.0914$
Single-Pulse Floor $\delta_{\min}(3) = 0.2028$
Regime 1: Feasible Single-Pulse ($T = 20$)
Regime 2: Feasible Sequential Isolated-Pulse
Regime 3: Infeasible via Isolated Pulses ($T \leq 20$)
Regime 4: Proved Infeasible (All Adaptive & Determ.)